SUNAY DAGLI

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Education

University of California, Berkeley - College of Engineering **May 2023 - May 2024**

M.S. Electrical Engineering & Computer Sciences (5th Year Accelerated M.S. Program)

- Thesis: "Offshore Wind and Tidal Energy Integration into Power System Linear Programming Capacity Model" (EECS-2024-84)
- Research Area: Mathematical Optimization, Deep Learning & Sustainable Energy Systems

University of California, Berkeley - College of Engineering **August 2019 - May 2023**

B.S. Electrical Engineering & Computer Sciences, B.S. Energy Engineering

- Relevant Coursework: Deep Reinforcement Learning, Optimization Models, Machine Learning, Artificial Intelligence, Efficient Algorithms and Intractable Problems, Computer Security, Databases, Computer Architecture, Electric Power Systems
- Extracurricular: Institute of Electrical and Electronics Engineers Student Branch (President), Berkeley Engineers and Mentors (Mentor), Certificate in Entrepreneurship & Technology

Experience

Software Engineer, Google **May 2024 - Present**

- Architect and scale core components of the Google Cloud Platform (GCP) API Proxy and Model Context Protocol (MCP) infrastructure at the network edge of cloud and AI systems.
- Engineer high-performance proxy extensions using Envoy and WebAssembly (WASM), isolating custom routing, security policy enforcement, and protocol translation while optimizing traffic routing where every millisecond of tail latency matters.

Undergraduate Research Assistant, Renewable and Appropriate Energy Laboratory **January 2022 - May 2024**

- Architected Python optimization modules for SWITCH, UC Berkeley's large-scale power-capacity linear programming model, simulating long-term grid decarbonization through offshore wind, marine, and tidal energy integration.
- Authored Master's Thesis evaluating spatial-temporal generation profiles, transmission bottlenecks, and storage co-optimization across multi-decade planning horizons.
- Co-authored technical deliverables for the U.S. Department of Energy (DOE), building an interactive data inventory and analytical pipeline mapping U.S. blue-economy power demands and marine energy resources.

Software Engineering M.S. Intern, Google **May 2023 - August 2023**

- Designed and implemented high-throughput batch execution APIs within the Google Cloud Apps / Workspace organization, consolidating fragmented RPC workflows into unified bulk operations.
- Reduced client-server round-trip overhead and improved API throughput for downstream Workspace integrations through request coalescing, asynchronous error handling, and rigorous load testing.

Software Engineering Intern, Google **May 2022 - August 2022**

- Engineered spatial optimization algorithms and Java-based backend RPC services to improve ride-share pickup point coverage and routing accuracy across high-traffic global airports.
- Analyzed geospatial telemetry and user pickup friction to dynamically rank and surface optimal staging zones, owning the entire development lifecycle from crafting comprehensive design documentation and reviews to production launch.

Undergraduate Research Assistant, UC Berkeley Hybrid Systems Laboratory **September 2021 - January 2022**

- Built machine learning and autonomous path-planning pipelines on the C3 AI platform to route solar-powered floating ocean platforms for gigaton-scale marine carbon sequestration.
- Integrated open-source oceanographic telemetry with Python control algorithms to harness surface currents and maximize energy-positive navigation trajectories.

Software Engineering (STEP) Intern, Google **May 2021 - August 2021**

- Built an internal command-line interface (CLI) and backend data pipeline connecting Google Cloud infrastructure APIs to automate virtual machine telemetry retrieval and centralize engineering debugging workflows.
- Developed Java and SQL query tooling paired with a web visualization dashboard, achieving comprehensive reliability through automated JUnit and end-to-end integration test suites.

Student Assistant & Software Engineer Intern, Lawrence Berkeley National Laboratory **May 2020 - May 2021**

- Led full-stack development of the HydroGEN Data Hub web search platform, unifying experimental and computational water-splitting materials datasets across 5 U.S. National Laboratories and 30+ DOE-funded projects.
- Engineered automated Python metadata parsers, a CKAN database query engine, and an interactive web GUI enabling researchers to ingest, validate, and query multi-institutional datasets.

Projects

InGameStats - Seamless GUI for Tracking Basketball Statistics

- Created Java GUI for basketball leagues to input real-time statistics and determine the best strategies to employ; deployed in two premier leagues in Southern California with over 20+ coaches.

Website for Masked Heroes Initiative Nonprofit

- As Chief Technology Officer for a nonprofit, developed a website using HTML/CSS/JavaScript, which enabled donations of 40,000+ masks through grassroots funding to combat COVID-19, featured by L.A. Times and congressmen.

Skills

Languages/Packages: Python, C++, Java, Go (Golang), C, SQL, JavaScript, HTML/CSS, MATLAB, Scheme, R, Envoy Proxy, WebAssembly (WASM), Model Context Protocol (MCP), TensorFlow, Keras, Pandas, NumPy, SciPy, React, Angular.js

Platforms/Tools: Google Cloud Platform (GCP), Distributed Systems, Cloud Infrastructure, High Availability Architecture, Performance Engineering, Linear Programming & Optimization, C3 AI, CKAN, AWS, Apache Spark, GitHub, Jupyter, Figma, ArcGIS